Basics of Testing Assignment

1. Explain the Principles of Testing

Software testing is the process of evaluating software to identify defects and verify whether it meets the specified requirements. There are seven principles of software testing.

1. Testing Shows the Presence of Defects

Testing can demonstrate that defects are present in the software. However, testing cannot prove that the software is completely free from defects.

2. Exhaustive Testing is Impossible

It is not possible to test every possible input, condition, and combination in a software application. Testing should be planned based on requirements, risks, and priorities.

3. Early Testing Saves Time and Cost

Testing should start as early as possible in the Software Development Life Cycle (SDLC). Identifying defects early helps reduce the cost and effort required to fix them.

4. Defect Clustering

A small number of modules or components usually contain most of the defects. Testers should give more attention to high-risk and complex areas.

5. Pesticide Paradox

Using the same test cases repeatedly may eventually fail to identify new defects. Therefore, test cases should be reviewed and updated regularly.

6. Testing is Context Dependent

Testing methods and techniques depend on the type, purpose, requirements, and risks of the software.

7. Absence-of-Errors is a Fallacy

Finding and fixing defects does not guarantee that the software will be successful. The software must also satisfy user requirements and business needs.

2. Difference Between QA and QC

QA (Quality Assurance) and QC (Quality Control) are two approaches used to ensure software quality.

QA (Quality Assurance)	QC (Quality Control)
Process-oriented	Product-oriented
Focuses on defect prevention	Focuses on defect detection
Improves development processes	Checks the quality of the product
Mainly a preventive activity	Mainly a corrective/detective activity
Focuses on how the product is developed	Focuses on whether the product is correct
Concerned with process quality	Concerned with product quality
Quality Assurance	Quality Control